

International Cybersecurity and Digital Forensics Academy (ICDFA)

HUSSEINI SULEIMAN

2025/FWSD/11478

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DATE: 24TH September, 2026

SBT-DF203: Basic Networking Skills for Digital Forensics 2025/FWSD/11478

LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

Three-Week Delivery Block 3/3 | 19-25 September 2026

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Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 9
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Historical CodeGate CTF file.xz/file capture supplied for offline analysis

1. Source Materials and Slide Alignment

- Decrypt Wireless Packets: IEEE 802.11 frames, WEP/RC4, IV reuse, Aircrack-ng key recovery, airdecap-ng decryption, file carving and optional passphrase analysis.
- The slide uses the historical CodeGate CTF 2015 Good Crypto capture and the known WEP40 key A4:3D:F6:F3:74.
- The assignment asks for host/server IP addresses, MAC addresses, manually extracted images/HTML and the router password where the optional CTF challenge is completed.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical wireless CTF capture protected by deprecated WEP40 has been supplied. You will preserve and decompress the evidence, identify IEEE 802.11 traffic, recover or validate the 40-bit key using Aircrack-ng, decrypt the capture offline, inventory network endpoints, carve transferred files and explain why WEP is cryptographically unsafe.

3. Learning Outcomes

- Identify management, control and data frames in an 802.11 capture.
- Explain WEP40 structure, RC4 keystream use, IV reuse and integrity limitations.
- Use Aircrack-ng on a supplied historical capture only.
- Decrypt WEP traffic using airdecap-ng and verify output files.
- Extract IP/MAC endpoints, protocols, images and HTML from decrypted traffic.
- Distinguish a WEP key from a human passphrase and document optional CTF password analysis.
- Recommend modern wireless security controls.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Do not capture, crack or connect to any live wireless network. This lab is exclusively offline and limited to the supplied historical training file.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Offline capture analysis.
Wireless tools	aircrack-ng suite	Recover/validate key and decrypt capture.
Packet tools	Wireshark and tshark	Inspect encrypted/decrypted frames.
Carving tools	foremost, binwalk, file, strings	Recover transferred objects.
Compression	xz-utils	Decompress file.xz.
Evidence	Historical CodeGate CTF capture	Authorized offline dataset.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

The screenshot shows a Kali Linux virtual machine running VMware Workstation. The terminal window displays the following commands and output:

```
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(root@kali)~[~]
# mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print

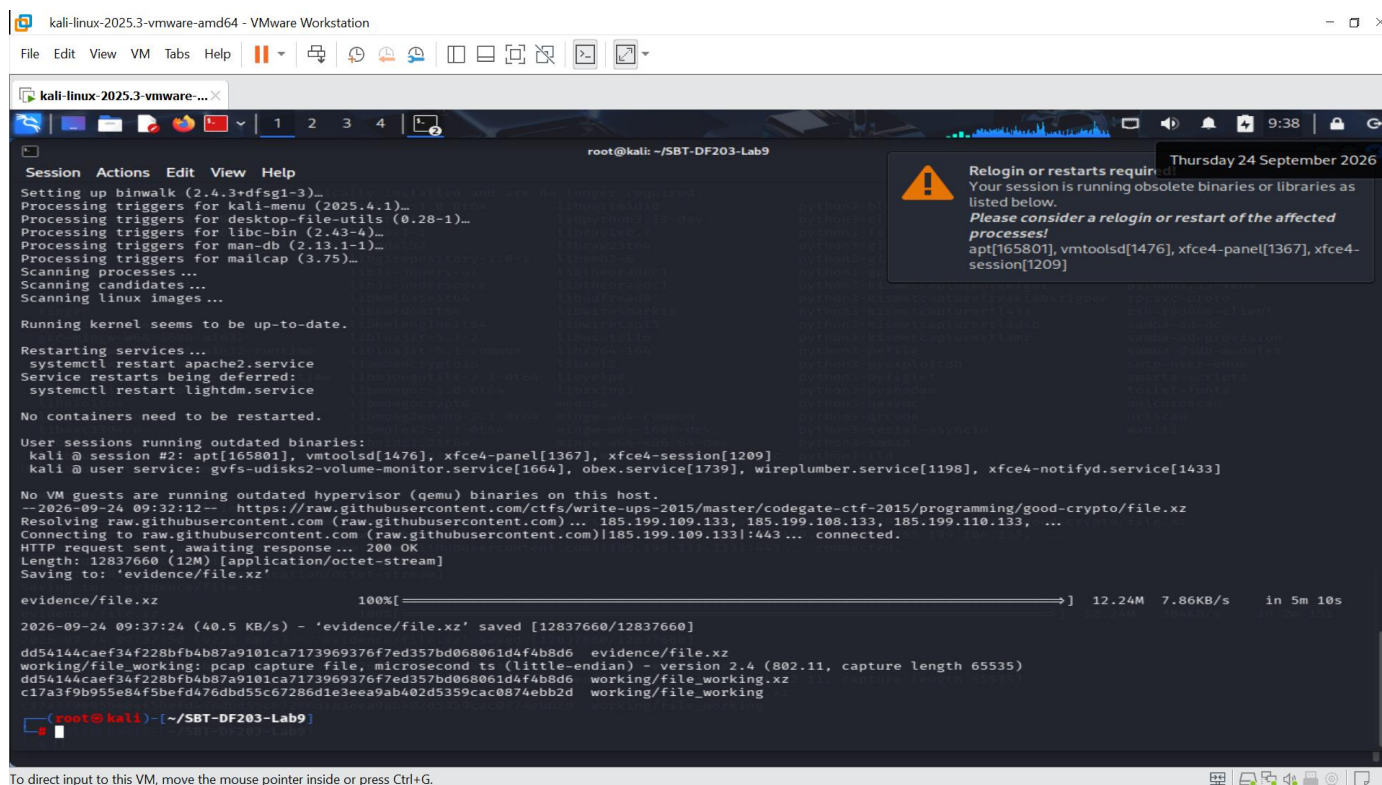
/root/SBT-DF203-Lab9
./exported
./evidence
./working
./screenshots
./scripts
./reports
```

A warning message from VMware is visible on the right side of the screen, indicating that the session is running obsolete libraries or plugins and suggesting a relin or restart.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
```

Screenshot required: Compressed evidence, decompressed capture type and SHA-256 hashes.



7. Mini Evidence and Chain-of-Custody Worksheet

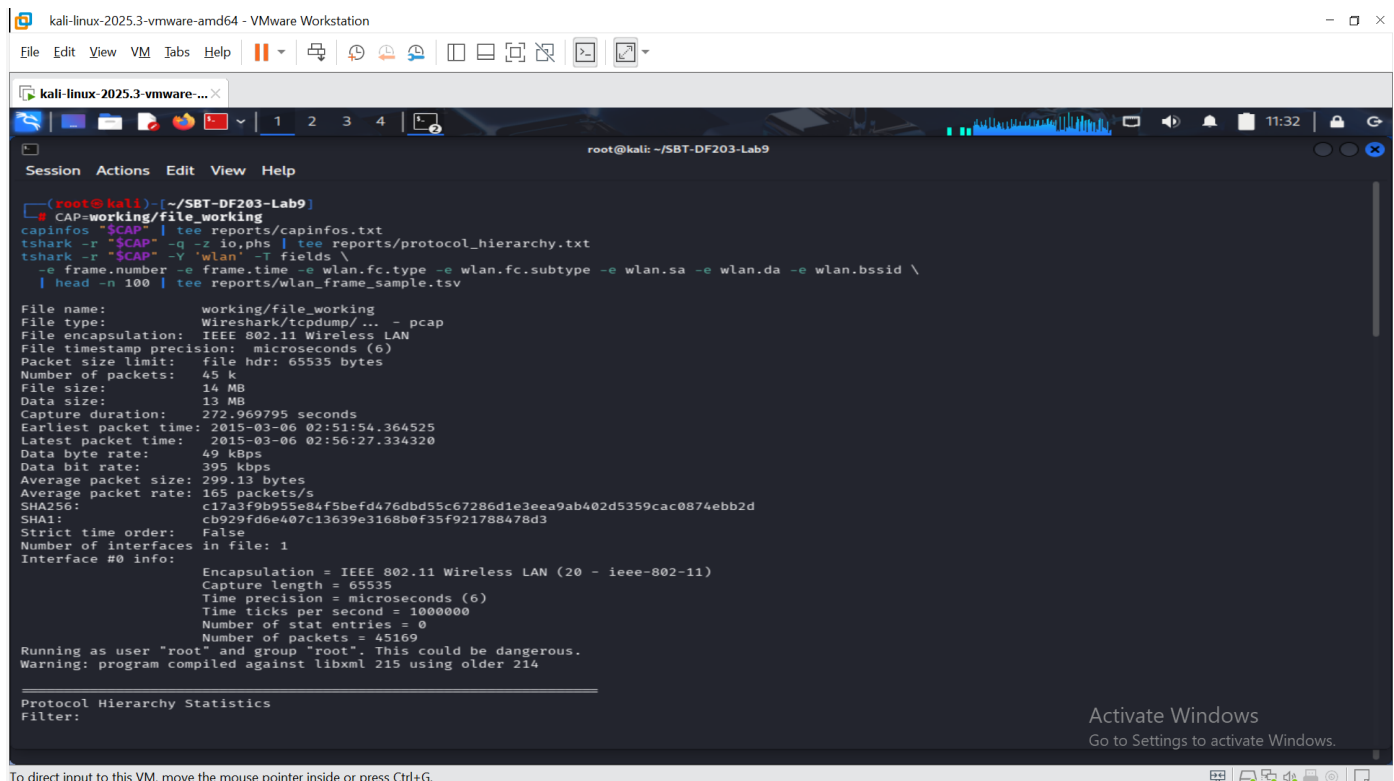
Field	Student Entry
Case/lab identifier	SBT-DF203-Lab9-Husseini Suleiman
Trainee name	Husseini Suleiman
Date and time started	24 th September, 2026
Evidence file name(s)	SBT-DF203-Lab9_2025FSWD11478_Husseini Suleiman.pdf
Source or generation method	Root@kali
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
Working-copy SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
Analysis workstation/VM	Hp elitBook/windows11/16gb/vm
Notes on any changes	nil

8. Part A - Inventory the Wireless Capture

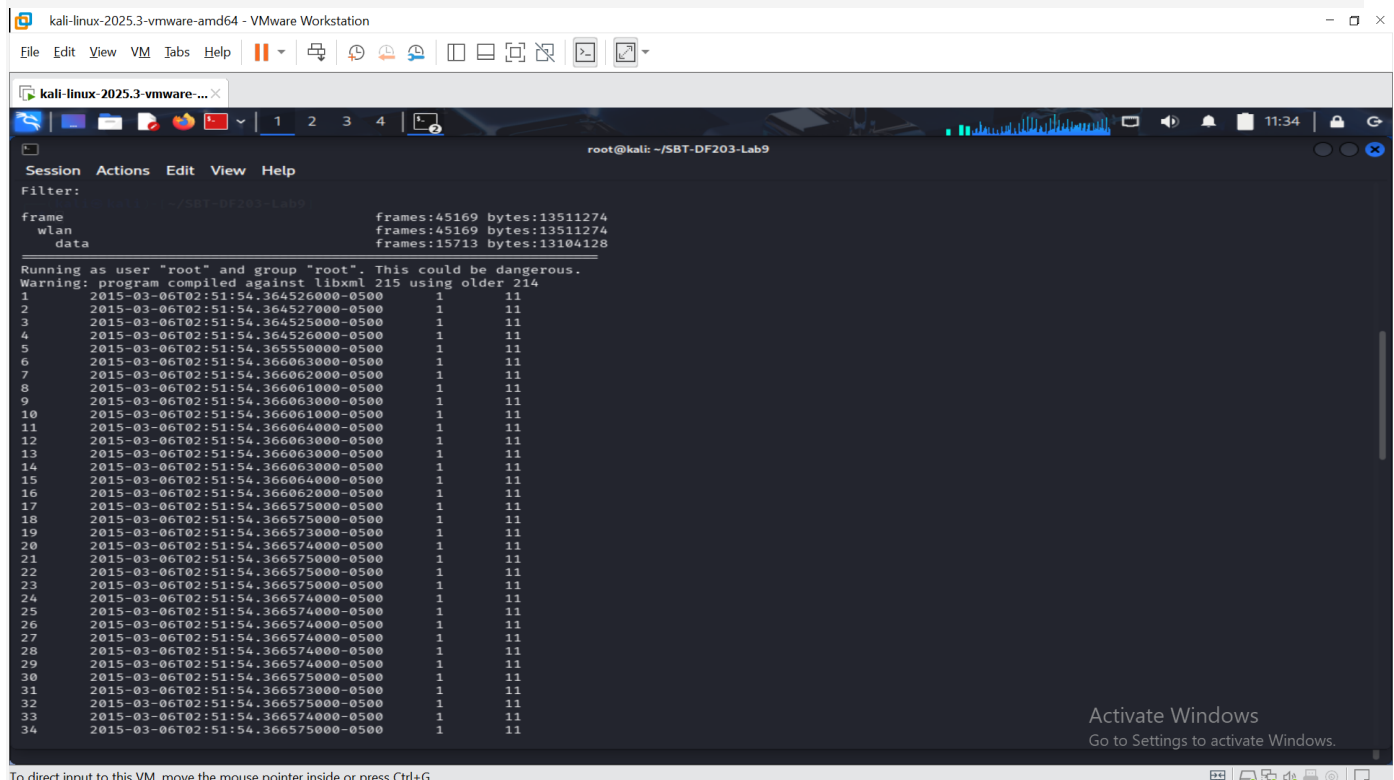
```

CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r "$CAP" -Y 'wlan' -T fields \
    -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
    | head -n 100 | tee reports/wlan_frame_sample.tsv

```



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
root@kali: ~/SBT-DF203-Lab9
Thursday 24 September 2026

Session  Actions  Edit  View  Help
43      2015-03-06T02:51:54.366574000-0500    1    11
44      2015-03-06T02:51:54.367085000-0500    1    11
45      2015-03-06T02:51:54.367086000-0500    1    11
46      2015-03-06T02:51:54.367087000-0500    1    11
47      2015-03-06T02:51:54.367088000-0500    1    11
48      2015-03-06T02:51:54.367089000-0500    1    11
49      2015-03-06T02:51:54.367090000-0500    1    11
50      2015-03-06T02:51:54.367091000-0500    1    11
51      2015-03-06T02:51:54.367092000-0500    1    11
52      2015-03-06T02:51:54.367093000-0500    1    11
53      2015-03-06T02:51:54.367094000-0500    1    11
54      2015-03-06T02:51:54.367633000-0500    0    8      00:26:66:55:97:d6      ff:ff:ff:ff:ff:ff      00:26:66:55:97:d6
55      2015-03-06T02:51:54.368111000-0500    1    11
56      2015-03-06T02:51:54.368112000-0500    1    11
57      2015-03-06T02:51:54.368113000-0500    1    11
58      2015-03-06T02:51:54.368114000-0500    1    11
59      2015-03-06T02:51:54.368115000-0500    1    11
60      2015-03-06T02:51:54.368116000-0500    1    11
61      2015-03-06T02:51:54.368622000-0500    1    11
62      2015-03-06T02:51:54.368623000-0500    1    11
63      2015-03-06T02:51:54.368624000-0500    1    11
64      2015-03-06T02:51:54.368625000-0500    1    11
65      2015-03-06T02:51:54.368626000-0500    1    11
66      2015-03-06T02:51:54.368627000-0500    1    11
67      2015-03-06T02:51:54.368628000-0500    1    11
68      2015-03-06T02:51:54.368629000-0500    1    11
69      2015-03-06T02:51:54.368630000-0500    1    11
70      2015-03-06T02:51:54.368631000-0500    1    11
71      2015-03-06T02:51:54.368632000-0500    1    11
72      2015-03-06T02:51:54.368633000-0500    1    11
73      2015-03-06T02:51:54.368634000-0500    1    11
74      2015-03-06T02:51:54.369134000-0500    1    11
75      2015-03-06T02:51:54.369135000-0500    1    11
76      2015-03-06T02:51:54.369136000-0500    1    11
77      2015-03-06T02:51:54.369648000-0500    1    11
78      2015-03-06T02:51:54.369649000-0500    1    11
79      2015-03-06T02:51:54.370159000-0500    1    11
80      2015-03-06T02:51:54.371182000-0500    1    11
81      2015-03-06T02:51:54.371183000-0500    1    11
82      2015-03-06T02:51:54.371184000-0500    1    11
83      2015-03-06T02:51:54.371694000-0500    1    11
84      2015-03-06T02:51:54.371695000-0500    1    11
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
root@kali: ~/SBT-DF203-Lab9
Thursday 24 September 2026

Session  Actions  Edit  View  Help
61      2015-03-06T02:51:54.368622000-0500    1    11
62      2015-03-06T02:51:54.368623000-0500    1    11
63      2015-03-06T02:51:54.368624000-0500    1    11
64      2015-03-06T02:51:54.368625000-0500    1    11
65      2015-03-06T02:51:54.368626000-0500    1    11
66      2015-03-06T02:51:54.368627000-0500    1    11
67      2015-03-06T02:51:54.368628000-0500    1    11
68      2015-03-06T02:51:54.368629000-0500    1    11
69      2015-03-06T02:51:54.368630000-0500    1    11
70      2015-03-06T02:51:54.368631000-0500    1    11
71      2015-03-06T02:51:54.368632000-0500    1    11
72      2015-03-06T02:51:54.368633000-0500    1    11
73      2015-03-06T02:51:54.368634000-0500    1    11
74      2015-03-06T02:51:54.369134000-0500    1    11
75      2015-03-06T02:51:54.369135000-0500    1    11
76      2015-03-06T02:51:54.369136000-0500    1    11
77      2015-03-06T02:51:54.369648000-0500    1    11
78      2015-03-06T02:51:54.369649000-0500    1    11
79      2015-03-06T02:51:54.370159000-0500    1    11
80      2015-03-06T02:51:54.371182000-0500    1    11
81      2015-03-06T02:51:54.371183000-0500    1    11
82      2015-03-06T02:51:54.371184000-0500    1    11
83      2015-03-06T02:51:54.371694000-0500    1    11
84      2015-03-06T02:51:54.371695000-0500    1    11
85      2015-03-06T02:51:54.372207000-0500    1    11
86      2015-03-06T02:51:54.372718000-0500    1    11
87      2015-03-06T02:51:54.375279000-0500    1    11
88      2015-03-06T02:51:54.375791000-0500    1    11
89      2015-03-06T02:51:54.376813000-0500    1    11
90      2015-03-06T02:51:54.376815000-0500    1    11
91      2015-03-06T02:51:54.376817000-0500    1    11
92      2015-03-06T02:51:54.376819000-0500    1    11
93      2015-03-06T02:51:54.376821000-0500    1    11
94      2015-03-06T02:51:54.376823000-0500    1    11
95      2015-03-06T02:51:54.376825000-0500    1    11
96      2015-03-06T02:51:54.381421000-0500    1    11
97      2015-03-06T02:51:54.381423000-0500    1    11
98      2015-03-06T02:51:54.382446000-0500    1    11
99      2015-03-06T02:51:54.382448000-0500    1    11
100     2015-03-06T02:51:54.382957000-0500    1    11
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Frame Category	Type Value	Purpose	Example Evidence
Management	0	Association, authentication, beacons	
Control	1	Flow control and acknowledgements	
Data	2	Carries network payload	

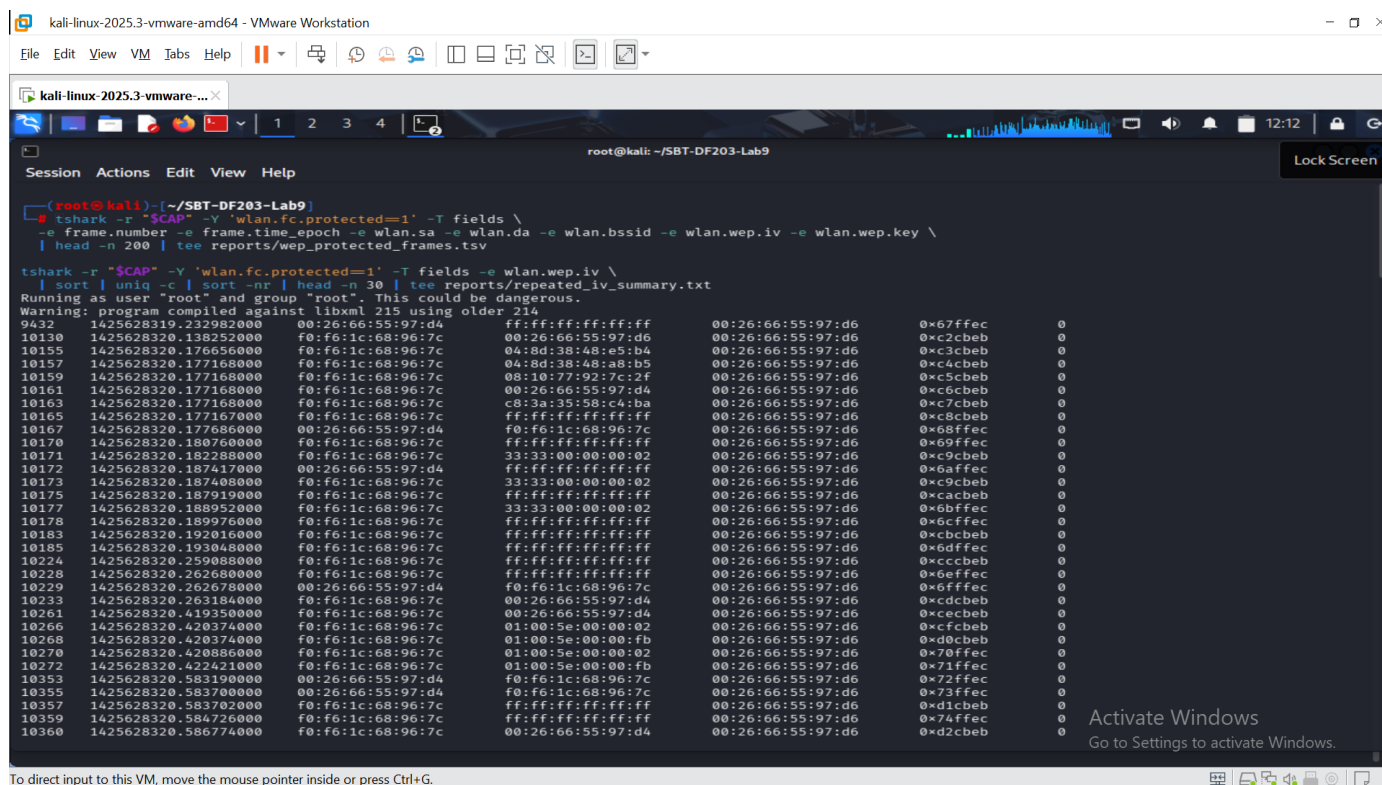
9. Part B - Identify WEP Protection and Key Parameters

1. Confirm that protected data frames are present.
2. Record BSSID, source/destination MAC addresses and IV-related fields where dissected.
3. Explain that 64-bit WEP normally combines a 24-bit IV with a 40-bit secret key.
4. Explain why repeated IVs and RC4 weaknesses make WEP unsuitable.

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv
```

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```

See screenshot of the above payload below



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

kali-linux-2025.3-vmware-... X

1 2 3 4

root@kali: ~/SBT-DF203-Lab9

Volume 53%

ES137/ES1373 / Creative Labs CT2518 (Audio PCI 64V/128/5200 / Creative CT4810/CT5803/CT5806 [Sound Blaster PCI]) Analog Stereo 7: d6 0x8cffee 0

Session	Actions	Edit	View	Help
10532	1425628323.220689000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
10534	1425628323.248340000	f0:f6:1c:68:96:7c	33:33:00:00:00:16	00:26:66:55:97:d6 0x8cffee 0
10536	1425628323.249874000	f0:f6:1c:68:96:7c	33:33:00:00:00:16	00:26:66:55:97:d6 0x8cffee 0
10537	1425628323.249877000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
10903	1425628324.976914000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6 0x8cffee 0
10905	1425628324.977937000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6 0x8cffee 0
10914	1425628326.256526000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10917	1425628326.258574000	f0:f6:1c:68:96:7c	33:33:00:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10919	1425628326.260115000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10920	1425628326.262162000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
10923	1425628327.256535000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10927	1425628327.256535000	f0:f6:1c:68:96:7c	33:33:00:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10929	1425628327.258068000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10930	1425628327.260116000	f0:f6:1c:68:96:7c	33:33:00:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
10972	1425628328.102932000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
10974	1425628328.103965000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
11031	1425628329.230422000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11033	1425628329.230422000	f0:f6:1c:68:96:7c	33:33:00:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11037	1425628329.238103000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11038	1425628329.240662000	f0:f6:1c:68:96:7c	33:33:00:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11045	1425628329.536596000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6 0x8cffee 0
11047	1425628329.537622000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6 0x8cffee 0
11192	1425628332.698386000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
11205	1425628332.874006000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11208	1425628332.902675000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
11215	1425628333.050710000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11218	1425628333.077333000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
11220	1425628333.080916000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0x8cffee 0
11222	1425628333.151574000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11228	1425628333.230420000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11230	1425628333.230421000	f0:f6:1c:68:96:7c	33:33:00:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11232	1425628333.231960000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6 0x8cffee 0
11241	1425628334.050710000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11243	1425628334.050710000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11245	1425628334.051222000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11247	1425628334.051222000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11249	1425628334.051734000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11251	1425628334.052246000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11253	1425628334.052247000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11255	1425628334.052759000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11257	1425628334.052759000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0
11259	1425628334.053271000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0x8cffee 0

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

kali-linux-2025.3-vmware-... X

1 2 3 4

root@kali: ~/SBT-DF203-Lab9

Lock Screen

Session	Actions	Edit	View	Help
11456	1425628340.818709000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0xcffec 0
11458	1425628340.819222000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0xcffec 0
11460	1425628340.819222000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0xcffec 0
11462	1425628340.819734000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0xcffec 0
11464	1425628340.819734000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0xcffec 0
11466	1425628340.820243000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6 0xcffec 0
11468	1425628340.821270000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6 0xcffec 0

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

7 0x42e8eb
7 0x28d3eb
6 0x9dfeb
5 0x9415ed
5 0x86e5eb
5 0x7ae5eb
5 0x28e7eb
5 0x19e5eb
4 0x45d7eb
4 0xd3e8eb
4 0xc915ed
4 0xc7d8eb
4 0xade4eb
4 0x73e5eb
4 0x65d8eb
4 0x4ad5eb
4 0x31e7eb
4 0x2723ed
4 0x10e7eb
4 0xffe5eb
3 0xf609ed
3 0xa307ed
3 0x97dbeb
3 0x96d1eb
3 0x92e8eb
3 0x91e8eb
3 0x8adeeb
3 0x630aed
3 0x56e8eb
3 0x3c1aed

(root@kali) - [~/SBT-DF203-Lab9]

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

10. Part C - Recover or Validate the WEP40 Key

Run Aircrack-ng against the supplied capture. Record selected network/BSSID, number of IVs, key length and recovered key. The slide-provided key is A4:3D:F6:F3:74, entered without colons for airdecap-ng.

```
aircrack-ng "$CAP" | tee reports/aircrack_output.txt
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
```

Evidence Handling

The recovered key is part of the historical CTF dataset, not authorization to attack any wireless network. Keep it inside the course submission.

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
root@kali:~/SBT-DF203-Lab9
# aircrack-ng "$CAP" | tee reports/aircrack_output.txt
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt

Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

# BSSID          ESSID          Encryption
1 00:26:66:55:97:D6  cgnetwork      WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0  0/ 1  A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944) 68(18944) 9E(18944) 0E(18688)
1  0/ 1  3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) 8D(19456) 80(19200) 85(19200) 9D(19200) 13(18944)
2  0/ 1  F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688)
3  1/ 10 F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200)
4  6/ 9  01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

A4:3D:F6:F3:74
root@kali:~/SBT-DF203-Lab9
Activate Windows
Go to Settings to activate Windows.
```

11. Part D - Decrypt the Capture Offline

```
cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt
```

Identify the file created by airdecap-ng, usually with a -dec suffix. Open it in Wireshark and confirm that IP/TCP/HTTP or other higher-layer protocols are now visible.

```
(root@kali) [~/SBT-DF203-Lab9]
# cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt

Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets  0
Number of plaintext data packets  0
Number of decrypted WEP packets  15477
Number of corrupted WEP packets  0
Number of decrypted WPA packets  0
Number of bad TKIP (WPA) packets  0
Number of bad CCMP (WPA) packets  0
524771 13904 -rw-rw-r-- 1 root root 14234002 Sep 24 09:37 working/file_working
524781 12488 -rw-rw-r-- 1 root root 12784501 Sep 24 12:16 working/file_working-dec
524770 12540 -rw-rw-r-- 1 root root 12837660 Sep 24 09:37 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz

(root@kali) [~/SBT-DF203-Lab9]
```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12. Part E - Extract Endpoints and Conversations

```
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
```

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

kali-linux-2025.3-vmware-... X

root@kali: ~/SBT-DF203-Lab9 CPU usage: 49.4%

```
Session Actions Edit View Help
root@kali:~/SBT-DF203-Lab9
# DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture.txt

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
  | head -n 200 | tee reports/ip_mac_mapping_sample.tsv

Decrypted capture: working/file_working-dec
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

Ethernet Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
Apple_68:96:7c: 15407 12 MB 7076 616 kB 8331 11 MB
EFMNetworks_55:97:d4 15020 12 MB 8178 11 MB 6842 593 kB
ASUSTekCOMPU_2a:c2:7a 358 281 kB 223 270 kB 135 11 kB
IPv4mcast_fb 42 6,648 bytes 0 0 bytes 42 6,648 bytes
IPv6mcast_fb 36 7,022 bytes 0 0 bytes 36 7,022 bytes
Broadcast 31 3,386 bytes 0 0 bytes 31 3,386 bytes
IPv4mcast_7f:ff:fa 31 11 kB 0 0 bytes 31 11 kB
IPv6mcast_16 10 920 bytes 0 0 bytes 10 920 bytes
IPv6mcast_02 9 630 bytes 0 0 bytes 9 630 bytes
IPv4mcast_16 3 180 bytes 0 0 bytes 3 180 bytes
IPv4mcast_02 2 92 bytes 0 0 bytes 2 92 bytes
NetisTechnol_48:e5:b4 1 42 bytes 0 0 bytes 1 42 bytes
NetisTechnol_48:a8:b5 1 42 bytes 0 0 bytes 1 42 bytes
08:10:77:92:7c:2f 1 42 bytes 0 0 bytes 1 42 bytes
TendaTechnol_58:c4:ba 1 42 bytes 0 0 bytes 1 42 bytes
IPv6mcast_ff:2a:c2:7a 1 78 bytes 0 0 bytes 1 78 bytes

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

IPv4 Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

kali-linux-2025.3-vmware-... X

root@kali: ~/SBT-DF203-Lab9

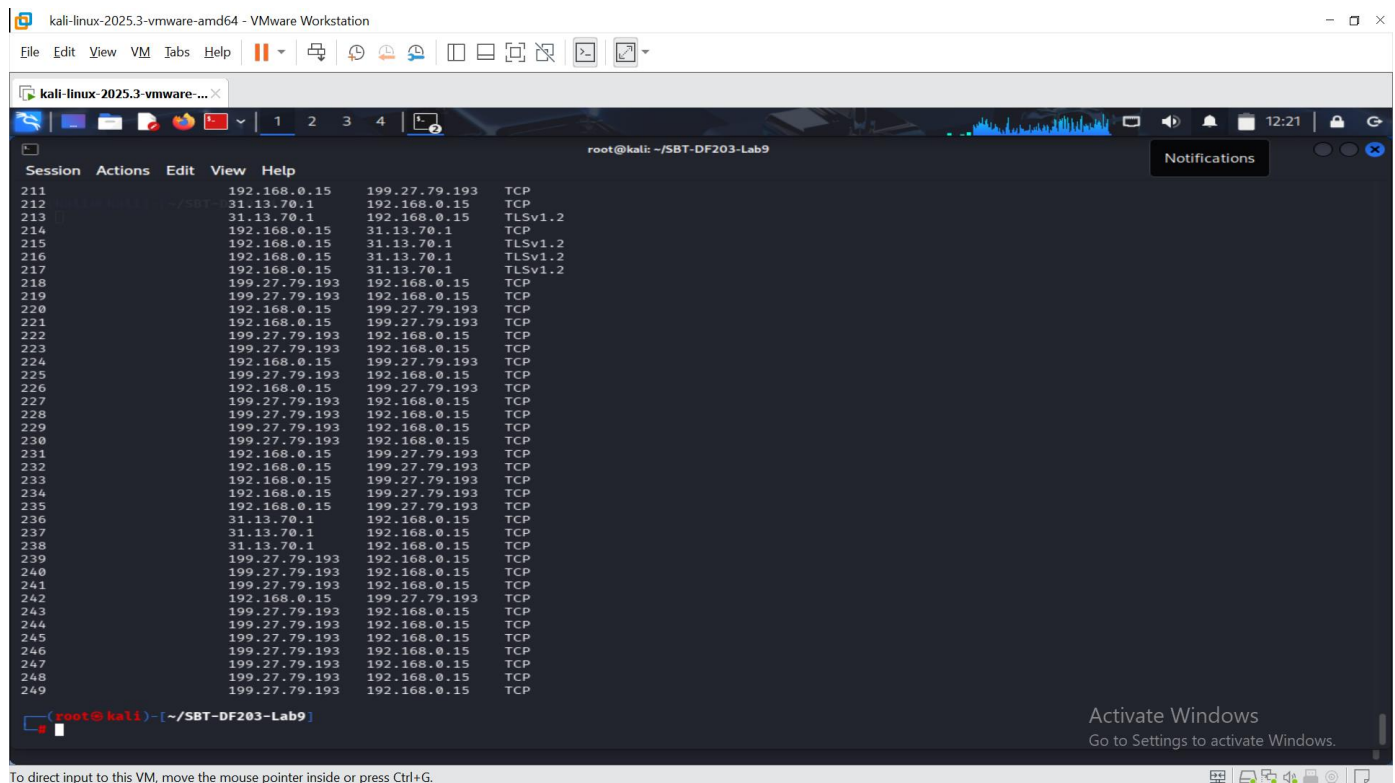
Computer
77% - 5 hours 15 minutes remaining

```
Session Actions Edit View Help
192.168.0.15:51011 ↔ 173.194.72.95:80 5 1,373 bytes 7 977 bytes 12 2,350 bytes 113.69112000 0.1363
192.168.0.15:51022 ↔ 50.31.251.34:80 6 2,777 bytes 6 706 bytes 12 3,483 bytes 153.704514000 5.0471
192.168.0.15:50972 ↔ 173.241.248.219:80 5 856 bytes 6 928 bytes 11 1,784 bytes 25.645117000 0.1347
192.168.0.15:50975 ↔ 173.241.248.219:80 5 964 bytes 6 933 bytes 11 1,897 bytes 25.753661000 0.1418
192.168.0.15:50976 ↔ 173.241.248.220:80 5 983 bytes 6 1,041 bytes 11 2,024 bytes 26.493053000 0.1424
192.168.0.15:50977 ↔ 173.241.248.220:80 5 948 bytes 6 1,046 bytes 11 1,994 bytes 26.721644000 0.6630
192.168.0.15:50978 ↔ 173.241.248.219:80 4 987 bytes 7 1,065 bytes 11 2,052 bytes 26.897084000 0.1377
192.168.0.15:50979 ↔ 173.241.248.220:80 5 699 bytes 6 817 bytes 11 1,516 bytes 27.298491000 0.1382
192.168.0.15:50981 ↔ 173.241.248.220:80 5 492 bytes 6 822 bytes 11 1,314 bytes 27.397819000 30.7886
192.168.0.15:51012 ↔ 173.194.72.95:80 4 859 bytes 7 868 bytes 11 1,727 bytes 113.691192000 17.0850
192.168.0.15:51023 ↔ 108.168.254.139:80 5 1,690 bytes 6 965 bytes 11 2,655 bytes 153.935492000 16.8300
192.168.0.15:51025 ↔ 108.168.254.138:80 5 1,124 bytes 6 1,209 bytes 11 2,333 bytes 155.335365000 31.6273
192.168.0.15:50965 ↔ 54.240.248.207:80 4 441 bytes 6 776 bytes 10 1,217 bytes 23.457211000 30.4936
192.168.0.15:50971 ↔ 173.194.127.100:80 4 715 bytes 6 1,206 bytes 10 1,921 bytes 25.457213000 30.3144
192.168.0.15:50982 ↔ 173.194.127.141:80 4 1,056 bytes 6 918 bytes 10 1,974 bytes 27.831546000 31.6820
192.168.0.15:50987 ↔ 54.169.42.244:80 4 1,054 bytes 6 1,000 bytes 10 2,054 bytes 28.157690000 30.8310
192.168.0.15:50994 ↔ 173.194.127.119:80 4 670 bytes 6 864 bytes 10 1,534 bytes 38.282619000 30.8592
192.168.0.15:50993 ↔ 173.194.127.128:80 3 604 bytes 6 864 bytes 9 1,468 bytes 38.275450000 31.1316
192.168.0.15:50957 ↔ 23.21.196.154:80 2 140 bytes 6 835 bytes 8 975 bytes 15.110590000 0.1403
192.168.0.15:50962 ↔ 203.190.124.19:80 3 648 bytes 5 1,094 bytes 8 1,742 bytes 24.496123000 0.0297
192.168.0.15:51037 ↔ 192.168.0.9:5000 3 1,162 bytes 4 631 bytes 7 1,793 bytes 243.326657000 0.3508
192.168.0.15:50955 ↔ 17.172.239.127:443 1 74 bytes 2 132 bytes 3 206 bytes 1.556832000 0.3759
192.168.0.15:50991 ↔ 173.194.122.15:80 1 74 bytes 2 132 bytes 3 206 bytes 31.473595000 0.3123
192.168.0.15:51024 ↔ 108.168.254.140:80 1 66 bytes 2 132 bytes 3 198 bytes 154.201220000 0.0512
192.168.0.15:51027 ↔ 173.194.127.226:443 1 74 bytes 2 132 bytes 3 206 bytes 172.165389000 0.1736
192.168.0.15:51031 ↔ 173.194.127.231:443 1 74 bytes 2 132 bytes 3 206 bytes 210.743425000 0.0246
17.172.232.98:443 ↔ 192.168.0.15:50948 1 54 bytes 1 103 bytes 2 157 bytes 16.865339000 0.1080
192.168.0.15:50988 ↔ 54.251.179.109:80 0 0 bytes 2 132 bytes 2 132 bytes 28.330747000 0.3820
192.168.0.15:50992 ↔ 173.194.122.23:80 0 0 bytes 2 132 bytes 2 132 bytes 31.674300000 0.0266
173.194.127.173:80 ↔ 192.168.0.15:50887 1 54 bytes 1 66 bytes 2 120 bytes 87.519168000 0.0261
173.194.127.173:80 ↔ 192.168.0.15:50889 1 54 bytes 1 66 bytes 2 120 bytes 87.677376000 0.0005
192.168.0.15:51029 ↔ 192.168.0.9:5000 1 60 bytes 1 78 bytes 2 138 bytes 181.299021000 0.0056
192.168.0.15:51033 ↔ 192.168.0.9:5000 1 60 bytes 1 78 bytes 2 138 bytes 211.516098000 0.0000
192.168.0.15:51030 ↔ 173.194.127.231:443 0 0 bytes 1 78 bytes 1 78 bytes 210.553984000 0.0000

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



13. Part F - Extract Images, HTML and Other Objects

5. First try Wireshark File > Export Objects for HTTP or other recognized protocols.
6. Use tshark object export for HTTP where available.
7. Use foremost as a carving method and document that carved files may include false positives or lack original filenames.
8. Open recovered objects only inside the lab VM and hash them before viewing.

```
mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
```

```
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
229 199.27.79.193 192.168.0.15 TCP
230 199.27.79.193 192.168.0.15 TCP
231 192.168.0.15 199.27.79.193 TCP
232 192.168.0.15 199.27.79.193 TCP
233 192.168.0.15 199.27.79.193 TCP
234 192.168.0.15 199.27.79.193 TCP
235 192.168.0.15 199.27.79.193 TCP
236 31.13.70.1 192.168.0.15 TCP
237 31.13.70.1 192.168.0.15 TCP
238 31.13.70.1 192.168.0.15 TCP
239 199.27.79.193 192.168.0.15 TCP
240 199.27.79.193 192.168.0.15 TCP
241 199.27.79.193 192.168.0.15 TCP
242 192.168.0.15 199.27.79.193 TCP
243 199.27.79.193 192.168.0.15 TCP
244 199.27.79.193 192.168.0.15 TCP
245 199.27.79.193 192.168.0.15 TCP
246 199.27.79.193 192.168.0.15 TCP
247 199.27.79.193 192.168.0.15 TCP
248 199.27.79.193 192.168.0.15 TCP
249 199.27.79.193 192.168.0.15 TCP

(root@kali) [~/SBT-DF203-Lab9]
# mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 0.000000 EFMNetworks_55:97:d4 -> Broadcast ARP 42 Who has 192.168.0.15? Tell 192.168.0.1
2 0.943674 Apple_68:96:7c -> NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.1? Tell 192.168.1.41
3 0.944186 Apple_68:96:7c -> NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
4 0.944186 Apple_68:96:7c -> EFMNetworks_55:97:d4 ARP 42 Who has 192.168.1.1? Tell 192.168.1.41
5 0.944186 Apple_68:96:7c -> EFMNetworks_55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
6 0.944186 Apple_68:96:7c -> TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10.1? Tell 192.168.10.113
7 0.944185 0.0.0.0 -> 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
8 0.944704 EFMNetworks_55:97:d4 -> Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
9 0.947778 0.0.0.0 -> 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
10 0.949306 fe80::1c0c:f6e8:fedb:8f8e -> ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
54 5d42025c1f6793c70cc31512f2d71b9bd8466d02609dfdc46882c02d3726f21a
055e154927bba6b4a204c7142fd3945a006bcb9b526e76e59f8dafeb0c405b
e1f955ae757c8b966c827a31bd3a30f658ced11f387f8ebf05ab3368629
AHI401eAGpIA5K5e4AO8AYCwHMOE1AB6ti-1e6adk-1099195366p-784, 685, 1034
meX3D14256086365is=300, 2508uc=5
95e43f1f5fdb8289bd5aa0de6bb39e35cd1b8324c825b12a3c1433d93322268
c4d24f6b27cc7ceea56fbc786bb1f486fdad9a1f998f760f76d1f4671e105c
5c7e53b08b84e97c174b5d3b18544cec351d7a010059b8936ae567d0d5ba2916
67841c636cd3f171696ce88b3fa10f3f7d060722e462af5f6f1c8758b81
4fbba9f1faef95f7caceba5708d359307683fba641b78653b897c8484061f88
974bd921fe2d8a0b997d9f8bc12a676ef49920663be48813fe60b15e65bc8fb
1855e8af2188864f04fe1e602db1841b0bc847268ed943d8fb31dc67c0e3b7a
c7d473663722fed9d8d587d1fd1d3ef2603af14860dcadbeac32705e3f0c8b7
b02ab544d4d4d91bc73183089db613f7cd4c954bc79a21dff4785c9280af45a0
bc519b1f4a7e4effda93b8d2f9eff4a2a753f69d4f563f561f953a1eeb6a8b3
3320a08508ac4dc175230508907642e8247fd07fe95799482949f4a60e26287
ae1eadea754a617805d616d9faac28cc8211a36421c11864d4152826ff03d3dc
1e81216e67645c4363da811ba786e4b6a6baa850e43765323ef58b5293e6d78
5c19b1f4a7e4effda93b8d2f9eff4a2a753f69d4f563f561f953a1eeb6a8b3
5c3a1e4440326f06fb2d0ae1484a382f9c69fd7f9692e2242e84a500b1e1b2
96a296d224f285c67bee92c30f8a309157f0daa35dc5b87e410b78630a09cfc7
cae63ae9f144308f2d5243eb9a57d26bba1261fc66007d06563db4c3d6488
1d11ac950ff1be55ca0e69a48731563f958af82925252ca88c96ca472b161be
02e357aeab1b284a9e6a170885744310191740a96d99bb189146b1c5c547ea79e
a0d3a0a0ff7dc3bf32d176f3c3dcdad67aba2867c4f4d1f7af6355d2cfc6c44f8
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
ef1955ae757c8b966c8324835031bd3a30f658ced11f387f8ebf05ab3368629
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
53f7e56783c414746a9f7c617357509560a26b7fe3135248640eac142d6e0f9a
3f9bed324e0a6c1735b498da9b1f38673e2051a5b33201e7713a91dd180029ba6
f891fa7ef7652a26a54638dd7db0aebcb6067d72ec42a2a636523a6371207dc
534c09b9b2d436a30c260eaa82679d2ac970d4962399a8da227ee07162a755
681fb2c21414bf8c7cedbbe1d57f9506424383bc89bed026e85ea6c85d7ccf
ebf4f635a17d10d6eb46ba680b70142419aa322f2f28001a036d311a22ee9d2a
792d47b4d20b4cbf0980011ca0b6792125c4e2ec5e2321bbab1e06ae09676
ddf5dda0e02b185e8bb0a7f7bee90be74a8a384cb4700658e76e754c8bf558
ee1c79d0b533ec5f90acfi199d84f8cd7c7e07acbe9dfeac65116b63687950d

exported/http_objects/portfolio-slideshow.min.js
exported/http_objects/accordion.min.js
exported/http_objects/activeviewk3fid=osdim6avi-BLA0YQV35VNmMMTY9QWknIgcCADA_50u-eGAABAB0
9850tos-1163,0,0,0,0,0mtos-1163,1163,116,com%2Fads%3Fpage%3Dinsidex%26main%3D1%26not%3D%26ti
meX3D14256086365is=300, 2508uc=5
exported/http_objects/object6684.application%2Fjavascript
exported/http_objects/collapsible-widgets.js
exported/http_objects/hit(3).json
exported/http_objects/ImageViewerInline.js%3F1425580383
exported/http_objects/Figure-2.png
exported/http_objects/object6496
exported/http_objects/twitter.png
exported/http_objects/6i=19612611nocookies=undefined
exported/http_objects/comment-reply.min.js
exported/http_objects/objectc111
exported/http_objects/collapsible-widgets.css
exported/http_objects/share-email.png
exported/http_objects/object107
exported/http_objects/object6491
exported/http_objects/ad53fpage=inside0main=16hot=8time=1425608636
exported/http_objects/og115PA.mp4
exported/foremost/audit.txt
exported/foremost/httm/00001639.htm
exported/foremost/httm/00024427.htm
exported/foremost/gif/00000406.gif
exported/foremost/gif/00003653.gif
exported/foremost/gif/00010876.gif
exported/foremost/gif/00003719.gif
exported/foremost/gif/00002450.gif
exported/foremost/gif/00001260.gif
exported/foremost/gif/00010456.gif
exported/foremost/jpg/00000104.jpg
exported/foremost/jpg/00004538.jpg
exported/foremost/png/00001831.png
exported/foremost/png/00024242.png
exported/foremost/png/00010214.png
exported/foremost/png/00020736.png
exported/foremost/png/00010671.png
exported/foremost/png/00001765.png
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

14. Part G - Optional Historical Passphrase Challenge

Optional and Instructor-Assessed

The slide discusses deriving a seed and cracking the CTF passphrase using supplied challenge information. This is not required for the core WEP decryption outcome unless assigned. Do not adapt the code to live networks.

- Differentiate the 40-bit WEP key from the human-readable router passphrase.
- Record the known key, derived seed, stated character constraints and challenge hash.
- Run only the instructor-provided findseed.py or crackWEP40 script against the historical dataset.
- Document the recovered passphrase in a protected appendix, not in public screenshots.

15. WEP Security Analysis

- WEP uses RC4 with a short 24-bit IV, allowing IV reuse in busy networks.
- The same keystream reuse leaks relationships between plaintexts and enables statistical key recovery.
- WEP provides weak integrity and no modern protection against replay or key-recovery attacks.
- Recommended replacement is WPA2-AES/CCMP or WPA3 with strong authentication and protected management frames where supported.
- Legacy WEP should be removed rather than strengthened with longer passphrases.

16. Required Findings Worksheet

Question	Finding
Capture format and duration	The capture format is tshark from 0.000000 to 15477
BSSID	BSSID :00:26:66:55:97:D6 ESSID:cgnetwork Encryption:WEP (15477 IVs)
Primary station MAC addresses	68:96:7c (Active client host) 55:97:d4 (Access Point / Router) 48:e5:b4 (Other local device) 48:a8:b5 (Other local device) 58:c4:ba (Other local device) 08:10:77:92:7c:2f
Protected frame count	
Repeated IV evidence	
Recovered/validated WEP40 key	
Decrypted capture filename/hash	KEY FOUND! [A4:3D:F6:F3:74] Decrypted correctly: 100%
Host IP and MAC	192.168.0.15 a4:83:e7:68:96:7c
Server IP and MAC	IP: 192.168.0.1 Mac: 00:26:66:55:97:d4

Question	Finding
Key protocols observed	ARP (Address Resolution Protocol) DHCP (Dynamic Host Configuration Protocol) ICMPv6 (Internet Control Message Protocol for IPv6) DNS (Domain Name System) IGMPv2 (Internet Group Management Protocol version 2) TCP
Recovered images/HTML	1d11ac950ff1be55ca0e69a487331563f958afd8295252ca88c96c4e72b161eb exported/foremost/htm/00001639.htm 02e357aea1b284a9e6a710885744310191740a96d99bb189146b1cbc547ea79e exported/foremost/htm/00024427.htm a0d3a0aff7dc3bf32d2176fc3dcda6e7aba2867c4f4d1f7af6355d2cfc6c44f8 exported/foremost/gif/00000406.gif 8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00003653.gif 8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00010876.gif 8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00003719.gif ef1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629 exported/foremost/gif/00002450.gif 8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00001260.gif 53f7e56783c414746a9fc3612637509560424617e3135248640eac142d6e0f94 exported/foremost/gif/00010456.gif 3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6 exported/foremost/jpg/00000104.jpg f891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc exported/foremost/jpg/00004538.jpg 534c09b9b2d436a30c260ead82679d2a2c970d4962399a8da227ee07162a755 exported/foremost/png/00001831.png 681fb2c21414bfb0c7cedbbe1d57f9506424033bc89beed026e85ea6c85d7ccf exported/foremost/png/00024242.png ebf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a exported/foremost/png/00010214.png 792477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676 exported/foremost/png/00020736.png ddf5dd4e0ef2b185e8bb0af7b6e90ebe74a84384cb4700658e76e754c8bfe550 exported/foremost/png/00010671.png ee1c79d0b533ec5ff0acf199d84fbc7dc7e07acbe9defcac65416b63687950d exported/foremost/png/00001765.png
Optional router passphrase	Protected appendix only
Security conclusion	

Required Screenshots Checklist

- ☐ Compressed evidence file and hash.
- ☐ Decompressed capture identification.
- ☐ Protocol hierarchy.
- ☐ 802.11 frame categories.

- ☐ Protected WEP frames and IV evidence.
- ☐ Aircrack-ng result.
- ☐ airdecap-ng decryption result.
- ☐ Decrypted IP/MAC endpoint inventory.
- ☐ Exported image or HTML objects.
- ☐ Hashes of decrypted capture and recovered files.

Student Report Template

9. Executive summary and authorization.
10. Evidence preparation and hashes.
11. 802.11/WEP identification.
12. Key-recovery or validation procedure.
13. Offline decryption.
14. Endpoint and conversation analysis.
15. Object extraction.
16. Optional passphrase challenge.
17. Security weaknesses and recommendations.
18. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab9_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Evidence handling	10	Historical dataset preserved and hashed.
802.11/WEP analysis	20	Correct frame types, protected frames, BSSID and IV explanation.
Key recovery/validation	15	Correct Aircrack-ng procedure and key documentation.
Decryption	20	Successful airdecap-ng output and verification.
Endpoint/object forensics	25	IP/MAC findings and recovered files with hashes.
Report quality	10	Clear security conclusion and screenshots.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
WEP40 composition	24-bit IV plus 40-bit secret key.
Slide-provided key	A4:3D:F6:F3:74.
airdecap format	Enter key as A43DF6F374.
Primary weakness	IV reuse and RC4-related key-recovery weaknesses.
Post-decryption evidence	Higher-layer protocols, endpoints and objects become analyzable.
Modern recommendation	WPA2-AES/CCMP or WPA3; retire WEP.

Command Reference Summary

Command	Purpose
aircrack-ng capture	Recover or validate wireless key.
airdecap-ng -w KEY capture	Decrypt WEP traffic.
wlan.fc.protected==1	Filter protected 802.11 frames.
wlan.wep.iv	WEP initialization vector field.
tshark --export-objects	Extract protocol objects.
foremost -i FILE -o DIR	Carve files from evidence.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.